

3. Methods

Simulating physical processes has traditionally been done sequentially; even during the modern age of hardware supporting parallel execution, using computers to calculate the evolution of physical phenomena has been sequential. Why haven't scientists just started doing things in parallel? Because of that pesky thing call *causality*; *the ball must go up before it can come down*. Because of this temporal dependence (spatial dependence has had its own workarounds such as the Barnes-Hut algorithm [2,3]), simulation of large-time-scale physics has thus taken a long time to execute. The Parareal algorithm (PA), and other parallel-in-time integration algorithms, have been developed in the last few decades [CN](#) to specifically address this issue. Many details and variations of the Parareal algorithm have been investigated to find and address issues such as stability [CN](#), convergence rates [CN](#), application to higher-order differential equations [CN](#). The main goal of this investigation is to contribute another variation: an implementation of the Parareal algorithm using methods from high-performance computing.

Before the PA can be implemented using these high-performance methods, the algorithm must be decomposed into its central components. The Parareal algorithm begins by partitioning a single IVP into several IVPs on smaller domains via an initial, inaccurate, “root” solution. Then each of the “subproblems” are solved using a sequential, accurate method on different threads at the same time. The final data for each of the subsolutions is then combined with the respective data of the root solution to yield a more accurate (i.e. “corrected”) root solution. This new root solution is then used to repeat the process until convergence. The interpretation of the PA in terms of these recursive subproblems makes the algorithm *almost* embarrassingly parallel; the corrections to the root solution need to be done sequentially. In addition to this structure, the algorithms being evaluated in parallel manifestly depend on simple arithmetic; because of this simplicity, the PA is well-suited to be evaluated on the GPU. Likewise, distributed methods can be combined with GPU evaluation for further parallelization for either a single model (taking advantage of the recursive nature of the PA) or a system of models.

Example: Consider the motion of a thrown ball just after it leaves the hand over the course of 8 seconds (of course ignoring air resistance). This is going to be simulated on a machine with 8 available threads. The root initial value problem for this physical scenario can be modeled by:

$$P = \left\{ \underbrace{\partial_t^2 \vec{r} = \vec{g}}_{\text{Acceleration}}, \quad \underbrace{\vec{r}(0) = \vec{r}_0, \vec{v}(0) = \vec{v}_0}_{\text{Initial values}}, \quad \underbrace{[0\text{s}, 8\text{s}]}_{\text{time span}} \right\}. \quad (1)$$

3.1. The Parareal Algorithm

The Parareal Algorithm aimed to solve the problem of physics taking too long to simulate; it didn't.

The main idea of the Parareal algorithm (PA) is to break up a single IVP into many smaller IVPs using some low-accuracy solution, solve those in parallel using high-accuracy methods, correct your initial solution using the sub-solutions, then make a new low-accuracy solution based on the corrected data, and repeat this process until the solution doesn't change. The end result of this procedure is a solution identical to one produced by directly using the high-accuracy method while potentially taking a less time [4]. Because the PA wraps traditional (sequential) solvers, it could be considered a “meta-” or “higher-order” method to solve IVPs.

3.1.1. Preparing the subproblems

Let the second-order initial value problem P be defined such that

$$P = \{\mathcal{L}(t, u, \partial_t u, \partial_t^2 u) = f(t), \quad u(0) = u_0, \partial_t u(0) = v_0, \quad [t_0, t_0 + \Delta t]\}, \quad (2)$$

and $D = [t_0, t_0 + \Delta t]$ is the closed time interval from t_0 to $t_0 + \Delta t$. The discretization N of P should be determined by the number of available threads N_t such that $N = mN_t$, for some positive integer m . The discretization should be chosen in this manner for maximum performance and efficiency; if $N = mN_t + r$, and $0 < r < N_t$, each thread will solve a subproblem m times until on the $m + 1$ iteration where only r threads would be active while $N_t - r$ threads idle (assuming all threads are synchronized). This type of optimization is sometimes referred to as “*flooding the threadpool*” to mitigate *thread starvation* CN.

With the discretization decided, partition the time domain D into subdomains D_p such that

$$D_p = \left[t_0 + \frac{p}{N} \Delta t, t_0 + \frac{p+1}{N} \Delta t \right] = [t_p, t_{p+1}]. \quad (3)$$

Use an fast integration method $\mathcal{G}_0$ (such as the Euler method) to compute initial root solutions $\{u_p^0\}_p$, $\{v_p^0\}_p$ defined such that

$$\{u_p^0\}_p = \{u_0^0, u_1^0, u_2^0, \dots, u_{N-1}^0\} \quad (4)$$

$$\{v_p^0\}_p = \{v_0^0, v_1^0, v_2^0, \dots, v_{N-1}^0\} \quad (5)$$

and $\{u_p^0, v_p^0\} = \mathcal{G}(\Delta t, u_{p-1}^0, v_{p-1}^0, \partial_t^2 u)$.

Subproblems P_p take the same from as in Eq. 2 (this also means subproblems are themselves, problems), but instead using initial conditions defined by those in the initial root solution. For example, the subproblem P_1 for the second ($p = 1$) subdomain, takes the form

$$P_1 = \{\mathcal{L}(t, u, \partial_t u, \partial_t^2 u) = f(t), \quad u(0) = u_1^0, \partial_t u(0) = v_1^0, \quad [t_p, t_{p+1}]\}. \quad (6)$$

Alg 1 shows the pseudocode of this process for a given IVP and integration algorithm i.e. “propagator”, resulting in root solutions and the collected subproblems. With these subproblems in hand, the PA continues to its next stage: propagating these problems in parallel.

Alg 1: PREPARE THE SUBPROBLEMS

INPUT: Second order initial value problem P , Coarse solver C
OUTPUT: Solution for P made from `pos_seq` and `vel_seq`, and array of subproblems `subproblems`
// choose discretization
 1 $N \leftarrow$ number of subproblems *// e.g. multiple of # of computer cores*
// partition the domain of P into N subdomains e.g. $[a, b] \rightarrow \{[a, c], [c, b]\}$
 2 `subdomains` $\leftarrow$ `partition(P.domain)`
// use coarse solver to get positions and velocities for the root problem
 3 `pos_seq, vel_seq` $\leftarrow$ `propagate(P, C)`
// create subproblems
 4 **for** i from 1 to $N - 1$
 5 `subdomain` $\leftarrow$ i -th domain partition `subdomains[i]`
 6 `pos0` $\leftarrow$ initial position for i -th subproblem `pos_seq[i]`
 7 `vel0` $\leftarrow$ initial velocity for i -th subproblem `vel_seq[i]`
 8 `subproblems[i]` $\leftarrow$ `ivp` on `subdomain` with initial values `pos0` and `vel0` for acceleration $P.\text{acc}$
 9 **return** Solution for root problem with `pos_seq` and `vel_seq`, and array of subproblems `subproblems`

Alg 1: Prepare the subproblems

Example: Given the example IVP (Eq. 6) and the available threads,

- 1) There should be 8 subproblems because there are 8 available threads.
- 2) Create the 8 time sub-domains $[0, 1], [1, 2], \dots, [7, 8]$.
- 3) Use the initial position and velocity to quickly solve the root problem via the Euler method to give a sequence of 8 total positions $\vec{r}_p^0$ and a sequence of 8 total velocities $\vec{v}_p^0$.
- 4) Use the calculated positions and velocities as initial positions and velocities to create 8 subproblems on the associated subdomains following the form

$$P_p = \{ \partial_t^2 \vec{r} = \vec{g}, \quad \vec{r}(0) = \vec{r}_p^0, \quad \vec{v}(0) = \vec{v}_p^0, \quad [t_p, t_{p+1}] \}. \quad (7)$$

The result of this process is shown in Figure 1.

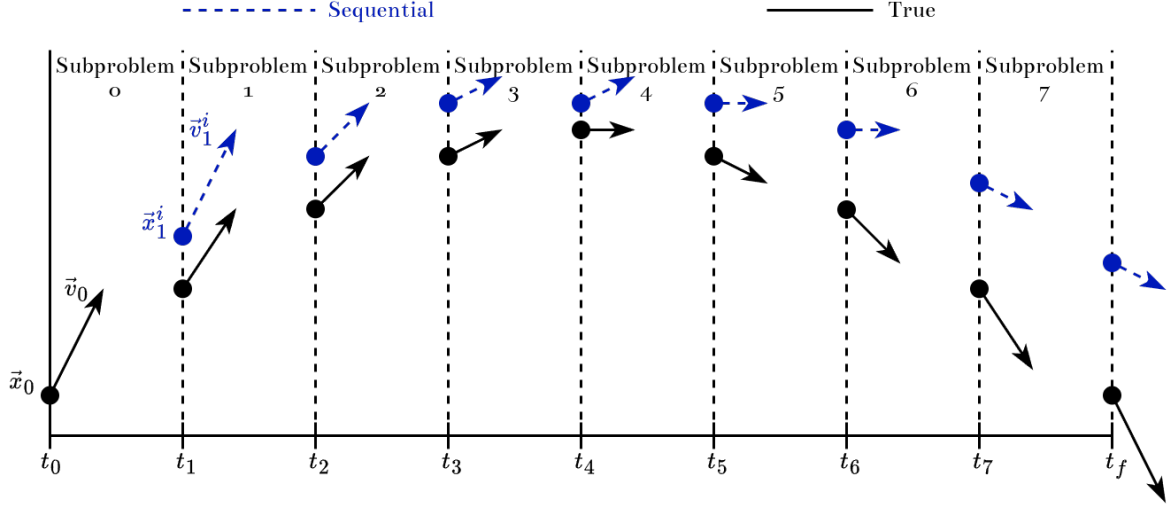


Figure 1: The motion of a ball flying through the air can be partitioned in time to form several initial value problems, each with its own initial position and velocity (upper, blue) determined by a fast integration method. Compared to the true solution (lower, black), this solution is very inaccurate.

3.1.2. Solving the subproblems

Discretizing the domain and propagating initial values as in the previous section constitutes the application of the *coarse propagator* $\mathcal{G}$ to the root problem. Because each of the produced subproblems is independent of the others, each can be accurately solved in parallel using a *fine propagator* $\mathcal{F}$ to reduce the total runtime by a factor equal to the number of subproblems. In other words, if applying $\mathcal{F}$ to a single subproblem has a runtime $\tau_{\mathcal{F}}$, then applying $\mathcal{F}$ to the root problem directly has a runtime $N * \tau_{\mathcal{F}}$ because there are N subproblems, whereas applying $\mathcal{F}$ in parallel only results in a runtime $\tau_{\mathcal{F}}$ because each application of $\mathcal{F}$ executes at the same time.

In general, a **propagator** $\mathcal{P}$ is defined by two key components: its integration algorithm $I_{\mathcal{P}}$, and its discretization $N_{\mathcal{P}}$. More specifically, the propagator $\mathcal{P}$ can be interpreted as a higher-order function mapping integration algorithms and discretizations to functions that, when applied to an IVP, reduce to sequences $\{(t, \tilde{r}_t)\}_t, \{(t, \tilde{v}_t)\}_t$ of time-position and time-velocity pairs, respectively; the collection of the sequences is called the **solution** S_P of P . The solution can be interpreted as the set of function-graphs (as defined in [5]) of $\tilde{r}$ and $\tilde{v}$, and is defined such that

$$\mathcal{P}(I_{\mathcal{P}}, N_{\mathcal{P}})(P) = \left\{ \{(t, \tilde{r}_t)\}_t, \{(t, \tilde{v}_t)\}_t \right\} =: S_P. \quad (8)$$

The application of the propagator to the subproblem is the core, or **kernel**, of the PA. While this description of the kernel is useful for understanding, it does not immediately lead to an algorithm that is well-suited for hardware-agnostic implementation (more details in section 3.2.1 on page 18). To that end, the implementation of the kernel presented here is composed of discretizing the subdomain and propagating the initial values separately.

Discretization: To avoid performance losses from each kernel allocating memory, each discretization kernel references a specific, pre-allocated, one-dimensional array δD of length N_p . In order for the kernel to generate the appropriate samplings of the domain, δD has its first and last elements pre-populated with the values of the lower and upper bounds of that thread's assigned subproblem such that

$$\delta D = \{t_p, [N_p - 2 \text{ arbitrary elements}], t_{p+1}\}. \quad (9)$$

With each kernel accessing this data, it can simply calculate and write the domain samples in-place; this process is shown in Alg 2.

Alg 2: PARALLEL DISCRETIZATION KERNEL

INPUT: Discretized domain `ddom` of length `N`
OUTPUT: Nothing

```

1 a, b  $\leftarrow$  (ddom[0], ddom[N-1])
2 step  $\leftarrow$  (b - a) / 2
3 for i from 1 to N - 2
4   ddom[i]  $\leftarrow$  a + (i - 1) * step
5 return
```

Alg 2: Each discretized subdomain is calculated by uniformly stepping from the lower bound to the upper bound. These results are written in-place.

Propagation: Much like the discretization kernel, the propagation kernel avoids memory allocations by using pre-allocated, pre-populated multidimensional arrays to store the position and velocity sequences; the initial values of the position and velocity for that kernel's subproblem are pre-populated as their respective arrays first element taking the form

$$\{\bar{r}_t\}_t = \{\bar{r}_{t_p}, [N_p - 1 \text{ arbitrary elements}]\} \quad \{\bar{v}_t\}_t = \{\bar{v}_{t_p}, [N_p - 1 \text{ arbitrary elements}]\} \quad (10)$$

Otherwise, the propagation kernel is no more than a traditional IVP solver as described in Section 2.1.2, but executed on many different subproblems simultaneously over many threads. This process is shown algorithmically in Alg 3.

Alg 3: PARALLEL PROPAGATION KERNEL

INPUT: A solver `solve`,
 acceleration function `acc`,
 sequence of `N` position vectors `pos_seq`,
 sequence of `N` velocity vectors `vel_seq`
OUTPUT: Nothing

```

1 for i from 2 to N - 1
2   old_pos, old_vel  $\leftarrow$  pos_seq[i - 1], vel_seq[i - 1]
3   pos_seq[i], vel_seq[i]  $\leftarrow$  solve(old_pos, old_vel, acc, step)
4 return
```

Alg 3: Each subproblem is solved in parallel using traditional methods.

The Parareal Kernel: In summary, the parareal kernel (Alg 4) is launched on each thread simultaneously and uses the fine propagator to populate arrays prepared from the domain and initial values of that thread's problem. The domain is discretized by uniformly stepping from the lower bound of the problem's domain to the upper bound. The initial values are propagated using a traditional integration method (Figure 2). The result of this process is sequences of times, positions, and velocities that solve that thread's problem for different discretizations.

Alg 4: THE PARAREAL KERNEL

INPUT: a fine propagator `fine`, discretized domain `ddom`,

position sequence `pos_seq`, velocity sequence `vel_seq`

OUTPUT: Nothing

1 Use `fine` in the discretization kernel to populate `ddom`

2 Use `fine` in the propagation kernel to populate `pos_seq` and `vel_seq`

3 **return**

Alg 4: Put it all together

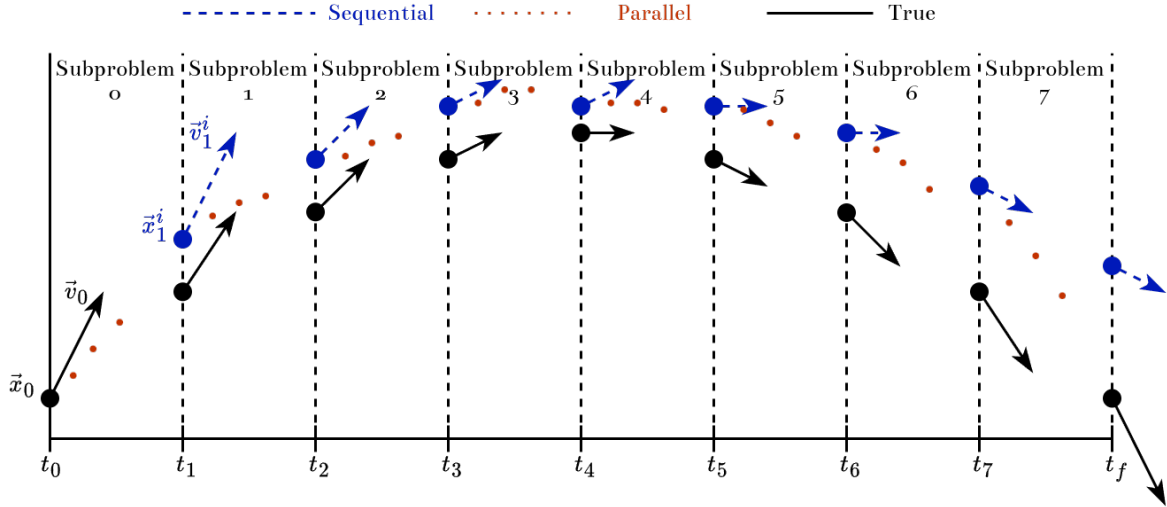


Figure 2: Each thread uses coarse and fine propagators to produce intermediate values (small, red dots) from the initial values (big, blue dots and arrows) of its assigned subproblem. Velocity data does exist, but is neglected here for visual clarity.

Example: For each of the 8 subproblems described by Eq. 7, each taking the form

$$P_p = \{\partial_t^2 \vec{r} = \vec{g}, \quad \vec{r}(0) = \vec{r}_p^0, \quad \vec{v}(0) = \vec{v}_p^0, \quad [t_p, t_{p+1}]\}.$$

- 1) Define the fine propagator $\mathcal{F}$ as the Velocity-Verlet method with a discretization of $N_{\mathcal{F}} = 2^6 = 64$.
- 2) Prepare arrays
 - 2.1) Allocate $3 * 8 = 24$ arrays of length 64 for the fine domains, positions, and velocities.
 - 2.2) Populate the beginning and end of each domain array with the lower and upper bounds, respectively, of each problem's domain.
 - 2.3) Populate the beginning of each position array with the initial position of each problem.
 - 2.4) Populate the beginning of each velocity array with the initial velocity of each problem.
- 3) Launch the Parareal Kernel with the propagators and prepared arrays
 - **Note:** If there are more problems than threads, the kernel can index stride [6]; more on this in Section 3.2.1.

3.1.3. Solving the root problem

Because the root solution has been calculated via an inaccurate method, it can be made more accurate using the results of the fine propagator. Though the root solution is not corrected only with the results of the fine propagator, but rather by coarsely propagating the root initial values again but adding a corrector determined by a combination of the results of the fine propagator and the previous iteration's root solution. The main idea of this process is known as *Deferred Corrections* [7].

The correction phase, as defined in literature [8], takes the deceptively-simple recursive form

$$u_t^i := \underbrace{\mathcal{G}(u_{t-1}^i)}_{\text{predictor}} + \underbrace{\mathcal{F}(u_{t-1}^{i-1}) - \mathcal{G}(u_{t-1}^{i-1})}_{\text{corrector}}, \quad (11)$$

where $u = \vec{r}, \vec{v}$ represents either position or velocity of the root problem. If the coarse propagation terms are collected as $\Delta_i \mathcal{G}_t^i := \mathcal{G}_t^i - \mathcal{G}_t^{i-1}$, Eq. 11 can be interpreted as *shooting method in time* [4]. It should also be noted that because the values returned by the coarse propagator are identical in successive iterations i.e. $i \rightarrow i+1 \Rightarrow \mathcal{G}(u_{t-1}^i) = \mathcal{G}(u_{t-1}^{i-1})$, they can be memoized and not calculated again (see Alg 5), leading to performance increases at the cost of storage space [9].

One way to interpret the correction equation is “at face value” as

On the current iteration, use the coarse propagator to recommend¹ what this value should be. Then correct it by adding the difference between the fine and coarse predictions from the previous iteration. Record this sum as the actual value.

An alternative interpretation arises from, effectively, “moving the correction to the top of the loop” as

Use the coarse propagator to traditionally evolve the root problem, but with an offset. This offset is initially zero.

¹This terminology is inspired by Giordano & Nakanishi's interpretation of the Gauss-Seidel and Simultaneous Over Relaxation methods in their seminal text on computational physics [10].

This interpretation changes “*use the propagator, then correct the value*” to “*correct the propagator, then use the value*”.

Thus the overall behavior of Eq. 11 could be defined through the explicit recurrence relation

$$\begin{aligned} u_t^i &= \mathcal{G}_t^i(u_{t-1}^i) \\ u_0^i &= u(0). \end{aligned} \tag{12}$$

Here the coarse propagator is effectively parameterized and can vary throughout iterations and times such that $\mathcal{G}_t^i(u_{t-1}^i) := \mathcal{G}(u_{t-1}^i) + \Delta_t^i$ and

$$\begin{aligned} \Delta_t^i &= \mathcal{F}(u_t^i) - \mathcal{G}(u_t^i) \\ \Delta_t^0 &= 0. \end{aligned} \tag{13}$$

Eq. 12 has the same structure of a traditional propagation making it much simpler to understand its behavior. This understanding is made even easier when considering Eq. 13 as it allows the Eq. 12 to also define the original coarse propagation of the root problem. In other words, the corrected coarse propagator has a correction of zero on the original iteration. While this interpretation is easier to understand, the result is mechanically identical to Eq. 11.

Alg 5: NEW SOLUTION GENERATOR

INPUT: Coarse solver `coarse`,
 Coarse position array `cpos`, Coarse velocity array `cvel`,
 Fine position array `fpos`, Fine velocity array `fvel`,
 Root position array `psol[i-1:i, :]`, Root velocity array `vsol[i-1:i, :]`
OUTPUT: Nothing
// evaluated at iteration i
 1 `psol[i, 0], vsol[i, 0] ← p0, v0`
 2 **for** `t` from 1 to `N-1`
 3 `ppred[i, t], vpredl[i, t] ← coarse(psol[i, t-1], vsol[i, t-1])`
 4 `psol[i, t] ← ppred[i, t] + fpos[i-1, t] - cpos[i-1, t]`
 5 `vsol[i, t] ← vpred[i, t] + fvel[i-1, t] - cvel[i-1, t]`

Alg 5: New position and velocity solutions are generated by coarse propagating the previous solution in the current iteration and combining it with the difference between the fine and coarse solutions from the previous iteration. The new solutions are pushed to a the end of the solution arrays.

Example:

- 1) Have arrays of positions and velocities at each time for the previous and current iteration; the first element of the current iteration’s array is the initial value of the problem, while the rest of empty. Also have arrays of positions and velocities generated from the coarse and fine propagators at each time for the previous iteration.
- 2) Use the new solution generator algorithm with these arrays and the coarse propagator.
- 3) The arrays for the position and velocity of the root solution at the current iteration are now populated.

3.1.4. Converging the root solution

Finally, as own in Alg 6, launch the parareal kernel (Alg 4) to gather the fine solutions for each point in time, and construct the new root solution (Alg 5) until the root solution stops changing between iterations. While there are many choices that can serve as valid convergence criteria [4], one of the simplest is:

$$\max_{1 \leq t \leq N-1} |u_t^i - u_t^{i-1}| < \varepsilon, \quad (14)$$

for some threshold ε . Eq. 14 determines convergence when every point in the solution changes by less than some amount between iterations.

Alg 6: THE PARAREAL ALGORITHM

INPUT: Root problem P , Coarse propagator G , Fine propagator F , Convergence threshold ε_p
OUTPUT: Discretized domain $\mathbf{ddom}$, Root position sequence $\mathbf{pos}$, Root velocity sequence $\mathbf{vel}$

```

1  $\mathbf{ddom}, \mathbf{pos}[0, :], \mathbf{vel}[0, :] \leftarrow$  Prepare the subproblems via a coarse solution
2  $i \leftarrow 1$ 
3 while  $\max(\mathbf{changes}) \geq \varepsilon_p$ 
4    $\mathbf{fpos}, \mathbf{fvel} \leftarrow$  Launch the parareal kernel with  $F$ ,  $\mathbf{pos}[i-1, :], \mathbf{vel}[i-1, :]$  to get the fine solutions
   to each subproblem
5    $\mathbf{pos}[i, :], \mathbf{vel}[i, :] \leftarrow$  Construct new root solutions with  $G$ ,  $\mathbf{pos}[i-1, :], \mathbf{vel}[i-1, :]$ , and
    $\mathbf{fpos}, \mathbf{fvel}$ 
6    $\mathbf{changes} \leftarrow$  The difference between the current and previous solutions
7 return  $\mathbf{ddom}, \mathbf{pos}, \mathbf{vel}$ 
```

Alg 6: The Parareal Algorithm is composed of looping two steps: finding the fine solutions in parallel, then finding the root solution sequentially. The loop stops when the root solution stops changing.

The magic of the Parareal algorithm lies in its divide-and-conquer approach to solving initial value problems. The “root” problem is sequentially and inaccurately solved to divide it into smaller problems whose initial values are defined by the solution. Those problems are simultaneously and accurately solved in parallel. The root problem is then solved in the same way as before, but at each step, the data is modified by combining the previous accurate and inaccurate solutions. Finally, the new root solution defines new problems, and the loop continues until the the solution has converged.